24—Q. How is the steam gage graduated to read?

A. Steam gages are graduated to read, pounds per square inch above the pressure of the atmosphere.

25—Q. What pressure is indicated by the steam gage?

A. The pressure per square inch inside the boiler.

26—Q. What shape is the tube in a Bourdon steam gage?

A. The tube is elliptical in section so that when the water is forced into it under pressure it will tend to become round and straighten out, thus moving the pointer to which it is attached.

27—Q. What qualities are determined in making a physical test of boiler steel?

A. In making a physical test of boiler steel, the qualities determined are the tensile strength, the elastic limit, the percent of elongation, and the percent of reduction of area at the point of fracture.

28—Q. On what principle does a steam gage work?

A. The steam gage pointer is actuated by a flattened or bent round tube to straighten itself under the pressure of steam against the water inside the tube. The gage pointer receives movement from suitable mechanism connected with the tube.

29—Q. What properties are required of rivets, braces, stays and staybolts?

A. Material for rivets, braces, stays and staybolts should possess the same qualities that are required of boiler plates.

30—Q. For what purpose are check valves used?

A. The purpose of a check valve is to allow water to flow through it into the boiler and to prevent the water and steam flowing back into the injector.

31—Q. What is meant by the efficiency of the joint, and how is it expressed?

A. The ratio between the strength of the joint and the strength of the plate is called the efficiency of the joint and is expressed as a percentage of the strength of the solid plate.

32—Q. Why are boiler checks placed so far from the firebox.

A. To introduce the water into the boiler at as great a distance from the firebox as possible. This permits the water to become heated to a higher temperature before it comes in contact with the firebox sheets.

33—Q. How is the efficiency of the joint determined?

A. In order to determine the efficiency of the joint its resistance must be computed for each of the different ways in which it may fail. Then if 100 times the lowest resistance of the several parts of the joint be divided by the resistance of the solid plate the efficiency of the joint will be obtained.

34—Q. For what purpose are water gage glasses and gage cocks used?

A. To indicate the height of water in the boiler.

35—Q. What is the difference in the efficiency of a triple riveted lap joint and a double riveted butt joint with two cover plates?

A. Triple riveted lap joints and double riveted butt joints with two cover plates have about the same efficiencies.

36—Q. In locating water gage cocks in a locomotive boiler where should the lowest one be placed?

A. The lowest water gage cock in a locomotive boiler should be placed not less than 3 inches above the highest part of the crown sheet.

37—Q. How does the angle in a crowfoot stay affect the stress on the stay?

A. The angle of the crowfoot stay increases the stress on the stay.

38—Q. Why is the level of the water in a gage glass the same as the level of the water in the boiler?

A. Because the lower fitting is connected with the water space and the upper fitting with the steam space.

39—Q. Define the term pitch as applied to rivets.

A. The distance from the center of one rivet hole to the center of the next rivet hole is called the pitch.

40—Q. What is the difference between a lifting and a non-lifting injector?

A. A lifting injector will create sufficient vacuum to raise the water from the level of the tank. The non-lifting injector will not raise the water, but merely force it into the boiler. A non-lifting injector must be placed below the level of the tank so that the water will flow to it by gravity.

41—Q. Is it customary to have inside laps face upwards or downwards?

A. Inside laps should always face downwards.

42—Q. What is a lifting injector?

A. A lifting injector is one that will lift the water to the height of the combining tube from the source that is not under pressure.

43—Q. What arrangement is made to give warning when a screw staybolt breaks?

A. In order to give warning when a screw staybolt breaks, a hole 3/16 inch in diameter is drilled in the outer end, not less than 1½ inches deep.

44—Q. What are essential parts of an injector?

A. The nozzles which force the water into the boiler, the operating mechanism, such as the lifting, steam and water valve.

45—Q. Will an injector work with compressed air?

A. No, an injector depends on the condensation of steam for its successful operation.

46—Q. What is the cause of laminations in a boiler plate?

A. Laminations are due to slag or other impurities in the metal.

47—Q. Upon what principle does an injector work?

A. Steam entering the injector at a high velocity carries the air inside the injector with it and creates a partial vacuum, causing the water to flow into it. The steam then imparts enough velocity to the water to give it sufficient momentum to throw open the check valves and enter the boiler.

48—Q. What causes foaming in a boiler?

A. Oil, alkali, or other matter will cause the water to foam.

50A—Q. Trace the path of the gases from the fire.

A. The gases of combustion after leaving the fire pass over the brick arch and through the flues to the smokebox, under the table plate and adjustable diaphragm, then they are drawn up through the petticoat pipe by the exhaust and through the stack to the atmosphere.

51—Q. What are some of the defects that will cause an injector to fail?

A. A leak in the feed pipe or tank hose or any leak between the injector and tank, insufficient water supply due to the tank valve partly closed, strainer stopped up, or tank hose kinked.

52—Q. How should a rigid staybolt be tested?

A. Would tap each bolt on the firebox side and determine the broken bolts from the sound or the vibration of the sheet.

53—Q. What is the maximum temperature at which a lifting injector can deliver the water in a boiler?

A. About 160 degrees F. or 71 degrees C.

54—Q. How should flexible staybolts without caps be tested?

A. Flexible staybolts without caps shall be tested the same as rigid bolts. Whenever a hydrostatic test is applied the staybolts shall be tested while the boiler is under hydrostatic pressure not less than the allowed working pressure.

55—Q. What is meant by an engine out of tram?

A. When corresponding wheels on opposite sides of the engine are not spaced equally apart, where the axle of any wheel is not at a right angle to the center line from front to rear of the engine so they do not run square on the rails, or when the space between the axle centers on opposite sides is not equal.

(To be continued)